

QIC 890: Quantum Complexity Theory

Problem Set 1

Due Friday, October 9, 2026

Problem 1

This problem consists of two parts. Taken together, solutions to the two parts imply that $\text{NP} \subseteq \text{PP}$.

- (a) Let $A = (A_{\text{yes}}, A_{\text{no}}) \in \text{NP}$. Prove that there exist a polynomial resource bound q and a language $C \in \text{P}$ such that the following implications are true.

$$x \in A_{\text{yes}} \implies |\{z \in \{0,1\}^q : \langle x, z \rangle \in C\}| > \frac{1}{2} \cdot 2^q \quad (1)$$

$$x \in A_{\text{no}} \implies |\{z \in \{0,1\}^q : \langle x, z \rangle \in C\}| \leq \frac{1}{2} \cdot 2^q \quad (2)$$

- (b) Prove that every promise problem $A = (A_{\text{yes}}, A_{\text{no}})$ for which there exist a polynomial resource bound q and a language $C \in \text{P}$ satisfying (1) and (2) is contained in PP (according to Definition 2.18 in the course notes).

Problem 2

Let $A \subseteq \{0,1\}^*$ be a language (which we may associate with the promise problem $(A, \{0,1\}^* \setminus A)$ as usual), assume $A \in \text{DTIME}(4^n)$, and define

$$B = \{xx : x \in A\}.$$

- (a) Give a high-level explanation (using words or pseudocode as you prefer) for why $B \in \text{DTIME}(2^n)$.
- (b) Prove that $A \leq_m^{\text{P}} B$.

Taken together, the two parts reveal that $\text{DTIME}(2^n)$ is not closed under Karp reductions. If it were, we would conclude that $\text{DTIME}(4^n) \subseteq \text{DTIME}(2^n)$, which violates the time-hierarchy theorem.

Problem 3

Section 5.1 of the course notes defines the *bounded-error quantum circuit acceptance* problem, which we abbreviate as QCIRCUIT-ACC, as follows:

BOUNDED-ERROR QUANTUM CIRCUIT ACCEPTANCE (QCIRCUIT-ACC)

Input The encoding of a quantum circuit Q over the standard gate set with no input qubits and one output qubit.

Yes $\Pr(Q \text{ accepts}) \geq 2/3$.

No $\Pr(Q \text{ accepts}) \leq 1/3$.

Prove that QCIRCUIT-ACC is BQP-complete with respect to Karp reductions.

Do not be concerned if your solution is straightforward — just focus on explaining this fact in as clear and simple terms as you can.

Problem 4

Define a complexity class PQP as follows. A promise problem $A = (A_{\text{yes}}, A_{\text{no}})$ is contained in PQP if there exists a polynomial-time uniform family $\{Q_x : x \in \Sigma^*\}$ of quantum circuits over the standard gate set, where each Q_x has no input qubits and one output qubit, such that these implications are true.

$$x \in A_{\text{yes}} \implies \Pr(Q_x \text{ accepts}) > 1/2$$

$$x \in A_{\text{no}} \implies \Pr(Q_x \text{ accepts}) < 1/2$$

In words, PQP is an unbounded-error version of BQP.

Prove that $\text{PQP} = \text{PP}$.

Suggestion: use results and possibly minor adaptations of arguments found in the course notes rather than starting from scratch. Both of the required containments $\text{PQP} \subseteq \text{PP}$ and $\text{PP} \subseteq \text{PQP}$ can be obtained in this way.